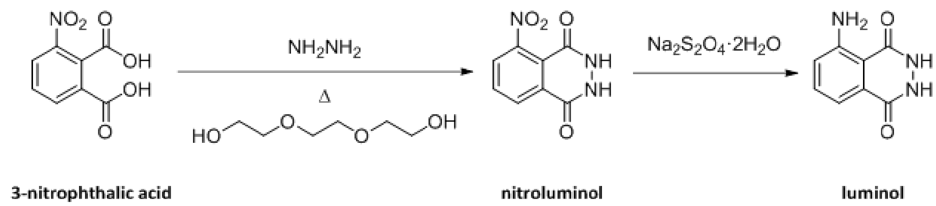
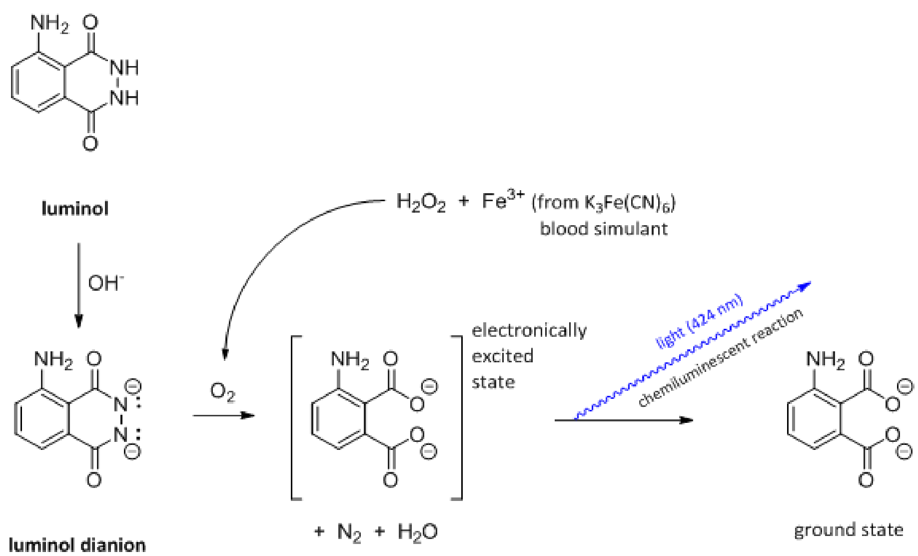
Activity 11: Forensic Chemistry: The Synthesis and Photophysical Properties of Luminol

The Experiment

Part 1. Two-Step Synthesis of Luminol from 3-Nitrophthalic Acid



Part 2. Photophysical Chemiluminescence of Luminol



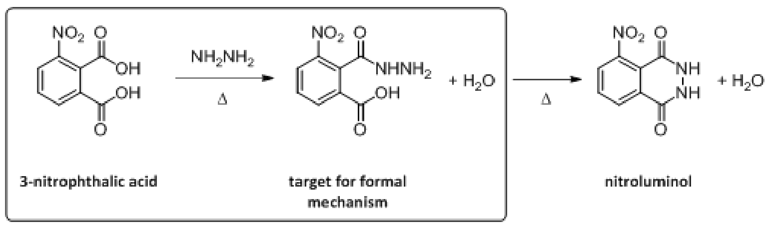
Lab Activity Goal

The goal of this week's lab is to synthesize luminol and test its reaction with a blood substitute. You will use an inorganic iron source to mimic an actual blood sample and hydrogen peroxide to produce the needed oxygen for this reaction.

Pre-Laboratory Assignments

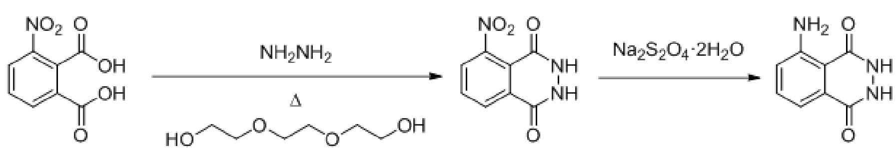
See the Activity 11 Pre-Lab Preparation and Outline document in Labflow.

ADDITIONALLY, draw into your notebook the mechanism for the portion of the reaction in the box below.



Experimental Procedure

Part 1. Synthesis of Luminol



Reaction 1: Preparing Nitroluminol

1. Heat 15 mL of water in an Erlenmeyer flask on a steam bath. Remember to add boiling stones before heating. This will be used step 8 of the procedure.
2. To a 50-mL round bottom flask add ~1 g of 3-nitrophthalic acid. Secure the flask (**no stir bar**) to the monkey bars. Lower flask into a heating mantle set on top of a raised lab jack. The bottom of the flask should have contact with the heating mantle ceramic.

Activity 11: Procedure

Important: Safety Pause

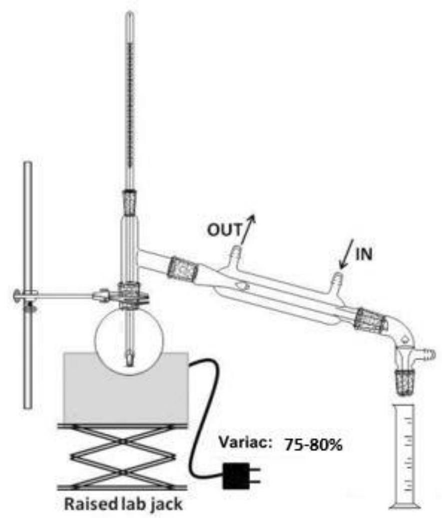
CAUTION: Hydrazine is corrosive. Wear gloves when handling the reagent and ensure it does not come into contact with your eyes, skin or clothing. Get new gloves once you are finished handling hydrazine.

Safety Pause

3. Take a 3-mL plastic syringe nestled in a 25-mL Erlenmeyer to the hydrazine (8%) dispensing station. Carefully draw up 2 mL of 8% hydrazine. Be sure you are wearing nitrile gloves. Place the loaded syringe in your Erlenmeyer, and carefully take the assembly back to your hood.
4. Carefully transfer hydrazine from your syringe into your round bottom flask that contains your nitrophthalic acid. Add a few boiling stones to the mixture and then use the Variac (75-80%) to gently heat the mixture until the solid acid dissolves.
   - When convenient, back at the hydrazine dispensing hood, rinse the syringe twice with DI water into the provided waste beaker before disposing of it into the solid waste.

5. Add 3 mL of triethylene glycol to the round bottom flask reaction solution.
6. Set up a simple distillation as depicted below taking into consideration the following points:
   - Check to make sure that the thermometer you are using is a high temperature thermometer (measures at least 200 °C).
   - The thermometer should be **submerged** in the reaction solution, but **not touching** the bottom of the flask.
   - **DO NOT** adjust the thermometer while it is inside the apparatus. Remove it, make a slight adjustment, and replace it. Repeat if necessary.

- **If the thermometer touches the bottom of the flask, it will break as the temperature rises.** If this occurs, you will not have to start over since the alcohol of the thermometer will not affect the remaining reactions.
- There is no need to save the water that distills out of the reaction vessel into the graduated cylinder.



7. Once your instructor inspects your apparatus, set the Variac to 80% and begin heating. When the thermometer temperature reaches 200 °C the reaction is considered complete.
8. Once your reaction has reached 200 °C, remove the flask from the heat source, cool to about 100 °C (crystals of the product often appear), and add the 15 mL of hot water that was heating on the steam bath.
9. Cool the reaction flask in an ice bath. Once it has nearly cooled to room temperature, collect the light yellow, granular nitro compound by vacuum filtration. The nitro compound does not need to be dried before proceeding to the next reaction.

Reaction 2: Reducing Nitroluminol to Luminol

10. Transfer the crude nitroluminol product from above back into the uncleaned flask in which it was prepared.

11. Securely clamp your flask and add a magnetic stir bar. While stirring, add 5 mL of 3 M NaOH solution. To the resulting deep brown-red solution add 3 g of loose powered sodium hydrosulfite dihydrate ($Na_2S_2O_4 \cdot 2H_2O$). Chunks may need to be broken up with a spatula. Wash the solid down the walls with a little water using a pipette as necessary.
12. After the additions are complete, add a heating mantle between the stir plate and the round bottom flask by lowering the lab jack enough to slide the mantle below the flask. While magnetically stirring, heat the reaction mixture to boiling at a Variac setting of **60-65%**. Once boiling, continue stirring for 5 minutes. During this time, some of the reduction product may separate.
13. Turn off the Variac, lower the lab jack, and remove the heating mantle.
14. Obtain 2 mL of glacial acetic acid according to the directions of your instructor.
15. Add 2 mL of glacial acetic acid to your reaction flask from step 12 above, cool the reaction flask on an ice bath, and stir. Collect the resulting precipitate of light yellow luminol by vacuum filtration.

Part 2. Chemiluminescence (Light-Producing) Reaction

Solution Preparation*:* Prepare each of the following solutions and set aside.

- **Solution 1 (blood mimic):** Into a 125 mL Erylenmeyer flask, add 4 mL of 3% aqueous potassium ferricyanide and 32 mL of water.
- **Solution 2 (luminol dianion):** Into a 50 mL beaker, add the moist luminol (dry weight about 40-60 mg), 2 mL of 3 M aqueous NaOH solution, and 18 mL of water. Gently swirl to dissolve solid.
- **Solution 3 (peroxide i.e. oxygen source):** Into a 50 mL beaker, add 4 mL of 3% hydrogen peroxide.

Chemiluminescence Experiment Set-Up: Team up with a complimentary group. One group (***Group A***) of the team will perform the chemiluminescence experiment at room temperature while the other group (***Group B***) of the team will perform the chemiluminescence experiment with ice bath-chilled solutions. **Each group of the team should commence at the same time** so that the effect of temperature on chemiluminescence can be directly compared.

Group A: Room Temperature Chemiluminescence

- At the same time as the complimentary chilled experiment (***Group B***), simultaneously pour solution 2 **AND** solution 3 into solution 1 (blood mimic).
- Observe the intensity and duration of light emission.

Group B: Low Temperature Chemiluminescence

- Prepare two ice baths.
  - Into one ice bath, place the 125 mL Erylenmeyer containing the blood mimic solution 1.
  - Into the other ice bath, carefully place the 50 mL beakers containing the luminol solution 2 and peroxide solution 3.
  - Let solutions chill for several minutes.
- At the same time as complimentary room temperature experiment (Group A), simultaneously pour the chilled solution 2 **AND** solution 3 into the chilled solution 1 (blood mimic).
- Observe the intensity and duration of light emission.

Part 3: Lab Check-Out

Before you leave lab, upload your in-lab notebook pages to Labflow. Your in-lab notebook pages must include the following in a **single pdf file**:

- Any experimental observations (including but not limited to: color changes, observational notes, measurements, sketches/images of your TLC plates, etc.)
- Your handwritten answers to the questions posted in the In-Lab Notebook Pages tile on Labflow
- A picture of your Organic Lab Cleaning Checklist

To complete your time in the organic lab, you will check out of your own locker and participate in part of a group clean-up of the lab:

1. **Locker Check Out**
   - Empty out your drawer
   - Remove and throw away the paper liner.
   - Get a new liner (either from your instructor or room 201).
   - Make sure ALL glassware is clean.
   - Ensure that drying tube is void of drying agent and cotton. The drying tube should also be rinsed.
   - Complete the check-out inventory on Labflow ("Organic Teaching Laboratory Locker Contents Check-Out" in the Activity 11 module).
   - Put any excess glassware that remains after you complete your inventory on Labflow in the back sink area. Be sure this excess glassware is clean.
   - Organize the contents of the drawer and have your instructor check off the items as you add them back into your drawer.
   - Have your instructor "sign" their name on the Labflow check-out as verification that you have followed the correct check-out procedure.
   - Lock your drawer

2. **Assigned Group Clean-Up Tasks**
   - Your lab instructor will tell you what your section's assigned clean-up tasks are. Follow their directions to keep our labs clean for future organic lab students. Thanks!

Clean-Up and Waste Disposal

Activity 11 Specific Clean-Up

- **Neutralization Stations (back right hood)**
  - Filtrates from steps 9 and 15
  - Chemiluminescence mixtures from part 2
  - Excess solutions 1, 2, and 3 from chemiluminescence experiment
  - Aqueous rinses from glassware washing
- **Special Hydrazine Waste (small front hood)**
  - Excess hydrazine
  - Water rinses from hydrazine syringes
- **Organic Liquid Waste**
  - Excess triethylene glycol
  - Acetone rinse from final glassware washing
- **Aqueous Waste**
  - Excess NaOH(aq) and acetic acid
- **Solid Waste**
  - Excess nitrophthalic acid
  - Rinsed syringe barrels (note that if our syringes had needles, the needles would need to go in a "sharps" waste)
  - Used filter paper
  - Used gloves
  - Used boiling stones
- **Any excess sodium hydrosulfite dihydrate must go into a specially labelled container in the back right hood.**

Routine Clean-Up

- Clean up your hood space of all trash, spilled chemicals, etc. Don’t leave a mess for the people who share this hood at other time slots. Up to 24 other people are sharing this space at various time slots throughout the week. Clean up your mess!

- No dirty or clean glassware should be left in the sink. No used filter paper or wadded up dirty paper towels should be left in the sink.
- **Dirty paper towels go into the REGULAR TRASH** not the red solid waste trash. This saves the university a lot of waste disposal money!
- Dispose of dirty filter paper in the red solid waste container.
    - Do **NOT** deposit filter paper in the liquid waste eco-funnel as this will cause a clog which will lead to a messy overflow of waste.
    - Do **NOT** leave filter paper in the sink as this is poor lab etiquette and rude.
- Used gloves should be disposed of into the red solid waste.
- **Ask your instructor if you are in charge of cleaning up the balance area** for this week. If you are in charge do the following:
    - Be sure all reagent bottles are properly capped.
    - Use the small brush to clean off balances.
    - Use small broom and dustpan (located on the wall surrounding the balance area) to remove excess solid from counter. Dispose of excess solids in red solid waste container. Return broom and dustpan to proper location.
    - Wipe down the counter with a moist paper towel.
- **Ask your instructor if you are in charge of making sure the sink area is presentable** for the next lab group.
    - Wash any dirty glassware rudely left in the sink. Let clean glassware air dry on rack or windowsill area.
    - Dispose of any paper towels, filter paper, or cotton rudely left in the sink.
    - **Do not remove broken glass from the drain or the sink**. Rather, tell your instructor and then fill out a "something wrong form" located in the cabinet above the solid waste containers. Be sure to record room number and the Left/Right location of the sink.

Post-Lab Assignment

There is no post-lab assessment for Activity 11.